Blockchain Exp - 3

Aim: Create a Cryptocurrency using Python and perform mining in the Blockchain created.

Theory:

1.Blockchain Overview

Blockchain is a **distributed and decentralized ledger** that stores information in a series of linked blocks.

Each block contains:

- Transaction data
- Timestamp
- Previous block's hash
- Its own unique hash (digital fingerprint)

Once data is recorded in a blockchain, it becomes **immutable** because altering one block would require recalculating all subsequent blocks.

2.Mining

Mining is the process of:

1. Collecting pending transactions into a block.
2. Performing a computational puzzle (Proof-of-Work) to find a valid hash.
3. Adding the new block to the blockchain. Broadcasting it to all connected peers.

Miners are rewarded with cryptocurrency for successfully mining a block.

3.Multi-Node Blockchain Network

In this lab, we simulate **three independent blockchain nodes** (5001, 5002, 5003). Each node:

- Runs on a separate port.
- Maintains its own copy of the blockchain.
- Can connect with peers to share and validate blocks.

4.Consensus Mechanism

We use the **Longest Chain Rule**:

- If multiple versions of the chain exist, the **longest valid chain** is chosen.
- This ensures all nodes agree on a single transaction history.

5.Transactions & Mining Reward

Each transaction has:

Sen•de

r

Rec•eiv

er

- Amount

When mining a block:

- Pending transactions are added to the block.
- A **reward transaction** is added automatically to pay the miner.

6.Chain Replacement

When /replace_chain is called:

1. Node requests chains from peers.
2. If it finds a longer and valid chain, it replaces its own.
3. This keeps the blockchain consistent across all nodes.

Tools & Libraries Used

- **Python 3.x**
- **Flask** – Web framework for API endpoints pip install Flask
- **Requests** – For HTTP communication between nodes pip install requests==2.18.4
- **Postman** – For testing API requests
- Python Standard Libraries:
  - datetime
  - jsonify
  - hashlib

- uuid4
- urlparse
- request

Code :

```
# Module 2 - Create a
Cryptocurrency # To be installed:
# Flask==0.12.2: pip install Flask==0.12.2
# Postman HTTP Client:
https://www.getpostman.com/ #
requests==2.18.4: pip install requests==2.18.4
# Importing the
libraries import
datetime import
hashlib import json
from flask import Flask, jsonify, request
import requests
from uuid import uuid4
hex from urllib.parse import urlparse                    # Generate a unique
                                                           id that is in # To
                                                           parse url of the
                                                           nodes

# Part 1 - Building a
Blockchain class Blockchain:
  def-init (self):
    self.chain =
    []
    self.transactions = []                        # Adding transactions before they
are added to a block
    self.create_block(proof = 1, previous_hash = '0')
    self.nodes = set()                            # Set is used as there is no
order to be maintained as the nodes can be from all around the globe

  def create_block(self, proof,
    previous_hash): block = {'index':
```

```
    len(self.chain) + 1,
        'timestamp':
        str(datetime.datetime.now()), 'proof':
        proof,
        'previous_hash': previous_hash,
        'transactions': self.transactions}              # Adding transactions to
make the blockchain a cryptocurrency
    self.transactions = []                          # The list of transacction
should become empty after they are added to a block
    self.chain.append(block)
    return block
```

```
  def
    get_previous_block(self):
    return self.chain[-1]

  def proof_of_work(self, previous_proof):
    new_proof = 1
    check_proof = False
    while check_proof is False:
      hash_operation = hashlib.sha256(str(new_proof**2 -
previous_proof**2).encode()).hexdigest()
      if hash_operation[:4] ==
        '0000': check_proof =
        True
      else:
        new_proof += 1
    return new_proof

  def hash(self, block):
    encoded_block = json.dumps(block, sort_keys =
    True).encode() return
    hashlib.sha256(encoded_block).hexdigest()

  def is_chain_valid(self, chain):
  previous_block = chain[0]
  block_index = 1
  while block_index < len(chain):
      block = chain[block_index]
      if block['previous_hash'] !=
        self.hash(previous_block): return False
      previous_proof =
      previous_block['proof'] proof =
      block['proof']
```

```
        hash_operation = hashlib.sha256(str(proof**2 -
        previous_proof**2).encode()).hexdigest() if hash_operation[:4] !=
        '0000':
            return False
        previous_block = block
        block_index += 1
    return True
```

- This method will add the trqnsaction to the list

```
of trnsactions def add_transaction(self, sender,
receiver, amount):
    self.transactions.append({'sender': sender,
                              'receiver':
                              receiver,
                              'amount':
                              amount})
    previous_block = self.get_previous_block()
```

```
        return previous_block['index'] + 1                    # It will return the block index
    to which the transaction should be
                added
```

- This function will add the node containing an address to the set of nodes created in init function

```
    def add_node(self, address):
        parsed_url = urlparse(address)                    # urlparse will parse the url
from the address
        self.nodes.add(parsed_url.netloc)                  # Add is used and not
append as it's a set. Netloc will only return '127.0.0.1:5000'
```

- Consensus Protocol. This function will replace all the shorter chain with the longer chain in all the nodes on the network

```
    def replace_chain(self):
        network = self.nodes                              # network variable is the set of
nodes all around the globe
        longest_chain = None                              # It will hold the longest chain
when we scan the network
        max_length = len(self.chain)                      # This will hold the length
of the chain held by the node that runs this function
        for node in network:
            response = requests.get(f'http://{node}/get_chain')    # Use get
chain method already created to get the length of the chain
            if response.status_code == 200:
                length = response.json()['length']            # Extract the length of the
chain from get_chain fiunction
                chain = response.json()['chain']
                if length > max_length and self.is_chain_valid(chain):       # We check if
the length is bigger and if the chain is valid then
                    max_length = length                                  longest_chain = chain
```

```
            # We update the max length                    # We update the longest chain
    if longest_chain:                          # If longest_chain is not none that
means it was replaced
        self.chain = longest_chain                  # Replace the chain of the
current node with the longest chain
        return True
    return False                               # Return false if current chain is the longest
one

# Part 2 - Mining our

Blockchain # Creating a

Web App
```

```
app = Flask( name )

#Creating an address for the node on Port 5000. We will create some other
nodes as well on different ports
node_address = str(uuid4()).replace('-', '')                #

#Creating a Blockchain
blockchain = Blockchain()

#Mining a new block
@app.route('/mine_block', methods =
['GET']) def mine_block():
  previous_block = blockchain.get_previous_block()
  previous_proof = previous_block['proof']
  proof =
  blockchain.proof_of_work(previous_proof)
  previous_hash = blockchain.hash(previous_block)
  blockchain.add_transaction(sender = node_address, receiver = 'Richard', amount =
1) #Hadcoins to mine the block (A Reward). So the node gives 1 hadcoin to Abcde
for mining the block
  block = blockchain.create_block(proof, previous_hash)
  response = {'message': 'Congratulations, you just mined a
        block!', 'index': block['index'],
        'timestamp':
        block['timestamp'], 'proof':
        block['proof'],
        'previous_hash': block['previous_hash'],
        'transactions': block['transactions']}
  return jsonify(response), 200

#Getting the full Blockchain
@app.route('/get_chain', methods =
```

```
['GET']) def_get_chain():
    response = {'chain': blockchain.chain,
            'length': len(blockchain.chain)}
   return jsonify(response), 200

#Checking if the Blockchain is valid
@app.route('/is_valid', methods =
['GET']) def is_valid():
   is_valid =
   blockchain.is_chain_valid(blockchain.chain) if
   is_valid:
     response = {'message': 'All good. The Blockchain is
   valid.'} else:
```

```
    response = {'message': 'Houston, we have a problem. The Blockchain is
  not valid.'} return jsonify(response), 200

# Adding a new transaction to the
Blockchain                                          # Post method as we
@app.route('/add_transaction', methods
= ['POST']) have to pass something to
get something in return def
add_transaction():

  json = request.get_json()                         # This will get the json file from
postman. In Postman we will create a json file in which we will pass the values
for the keys in the json file
  transaction_keys = ['sender', 'receiver', 'amount']
  if not all(key in json for key in transaction_keys):      # Checking if all
keys are available in json
    return 'Some elements of the transaction are missing', 400
  index = blockchain.add_transaction(json['sender'], json['receiver'],
  json['amount']) response = {'message': f'This transaction will be added
  to Block {index}'}
  return jsonify(response), 201                     # Code 201 for

creation # Part 3 - Decentralizing our Blockchain

# Connecting new nodes                              400 for node in
@app.route('/connect_node', methods =               nodes:
['POST']) register the new nodes from the             blockchain.add_node(node)
json file
def connect_node():
  json = request.get_json()
  nodes = json.get('nodes')
  file if nodes is None:
    return "No node",
```

```
            # POST request to                          # Get the nodes from json


  response = {'message': 'All the nodes are now connected. The Hadcoin Blockchain
now contains the following nodes:',
          'total_nodes':
  list(blockchain.nodes)} return
  jsonify(response), 201

# Replacing the chain by the longest chain if
needed @app.route('/replace_chain', methods =
['GET'])
def replace_chain():
  is_chain_replaced =
  blockchain.replace_chain() if
  is_chain_replaced:
    response = {'message': 'The nodes had different chains so the chain was replaced
by the longest one.',
```

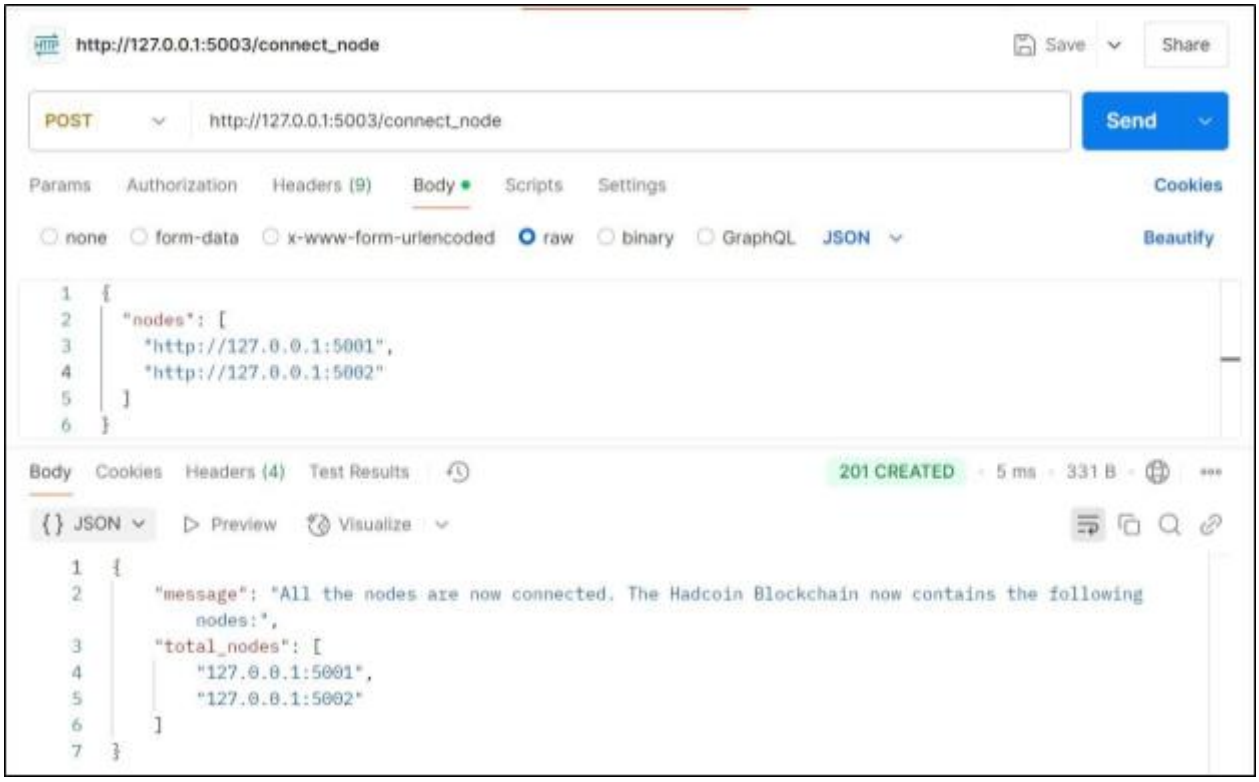
```
            'new_chain': blockchain.chain}
    else:
        response = {'message': 'All good. The chain is the
            largest one.', 'actual_chain': blockchain.chain}
    return jsonify(response), 200

# Running the app
app.run(host = '0.0.0.0', port = 500*)
```

Output :

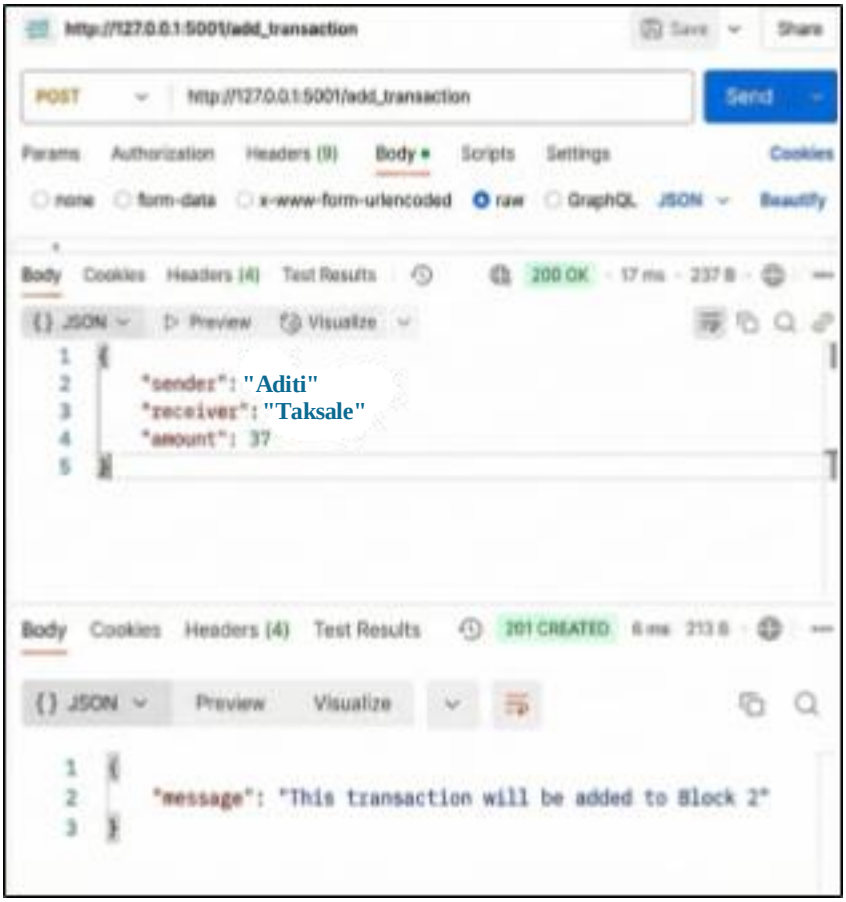
1.Connecting nodes: Node 1 -> 2,3

Node 2 -> 1,3

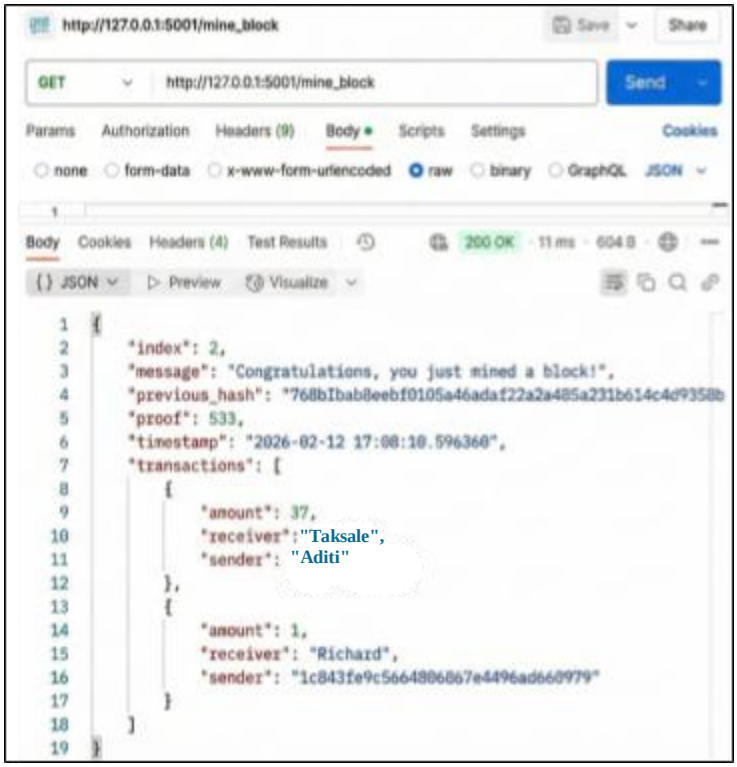
Node 3 -> 1,2



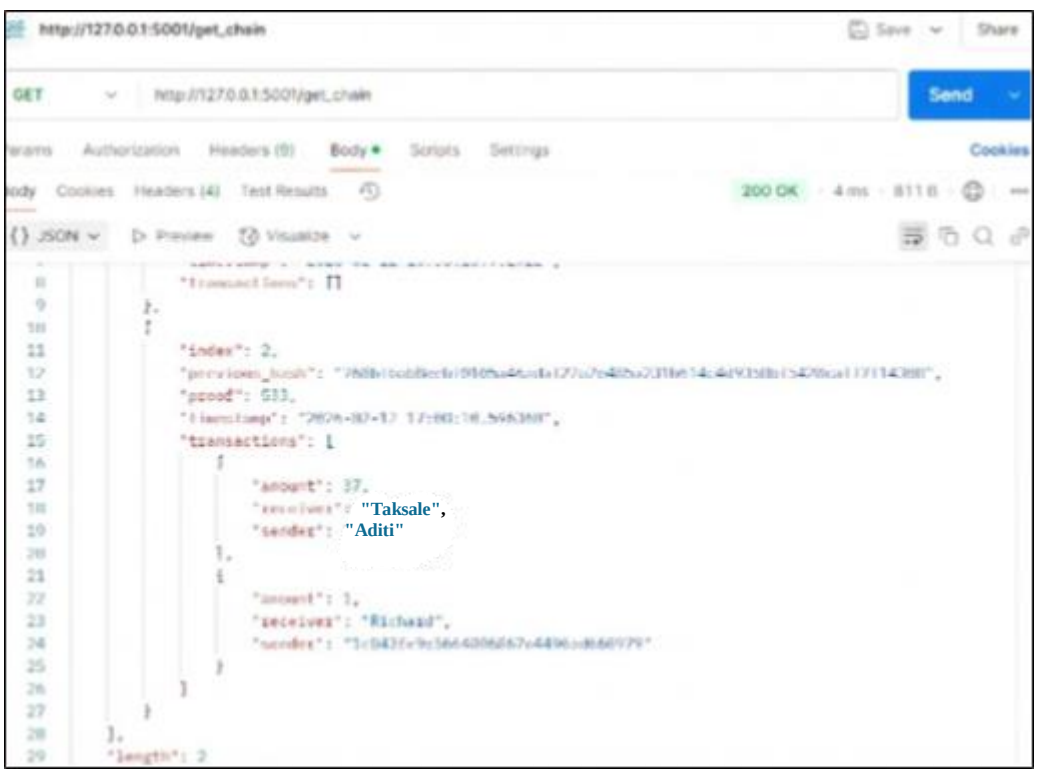
2. *Adding transaction from node 1*


http://127.0.0.1:5001/add_transaction
Save
Share
POST
http://127.0.0.1:5001/add_transaction
Send
Params
Authorization
Headers (9)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
GraphQL
JSON
Beautify
Body
Cookies
Headers (4)
Test Results
200 OK
17 ms
237 B
JSON
Preview
Visualize
{
"sender": "Aditi"
"receiver": "Taksale"
"amount": 37
}
Body
Cookies
Headers (4)
Test Results
201 CREATED
6 ms
213 B
JSON
Preview
Visualize
{
"message": "This transaction will be added to Block 2"
}

3.Mining a Block (Only on 5001)



4. Checking chain length difference: Node 1



Node 2

http://127.0.0.1:5002/get_chain

Save Share

GET http://127.0.0.1:5002/get_chain Send

Params Authorization Headers (9) Body Scripts Settings Cookies

Body Cookies Headers (4) Test Results 200 OK 4 ms 339 B

{} JSON Preview Visualize


```
{
    "chain": [
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-02-12 16:35:01.448893",
            "transactions": []
        }
    ],
    "length": 1
}
```


Node 3

http://127.0.0.1:5003/get_chain

Save Share

GET http://127.0.0.1:5003/get_chain Send

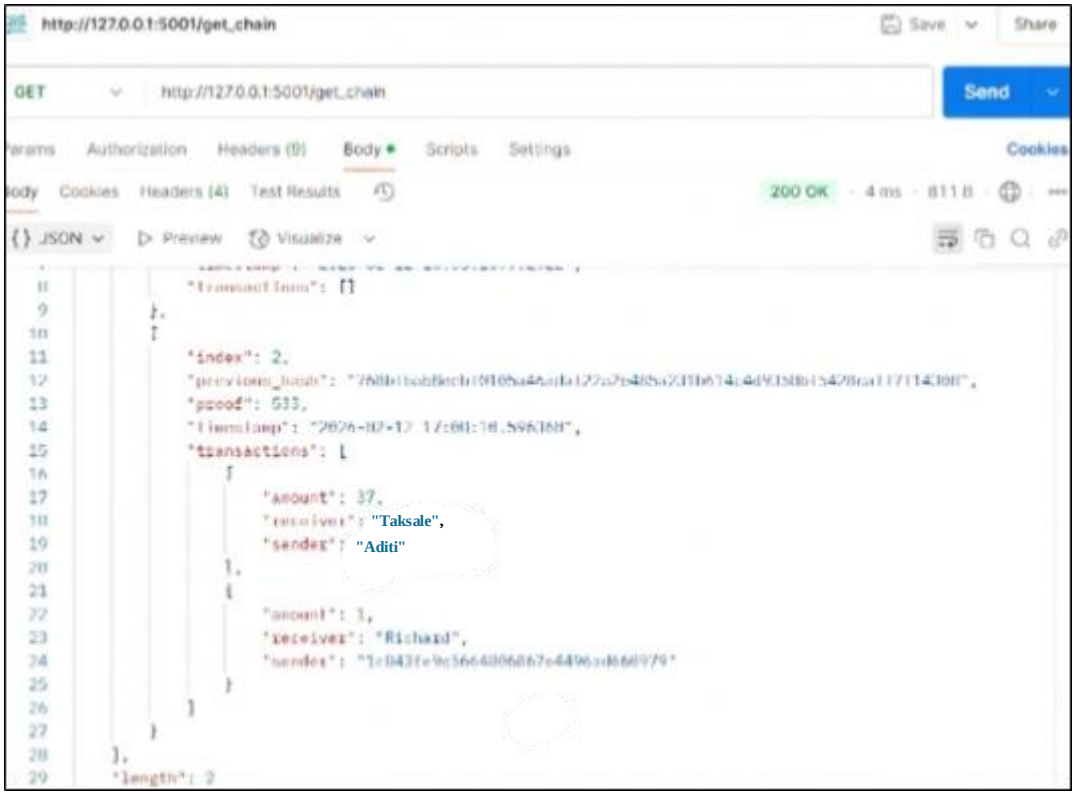
Params Authorization Headers (9) Body Scripts Settings Cookies

Body Cookies Headers (4) Test Results 200 OK 5 ms 339 B

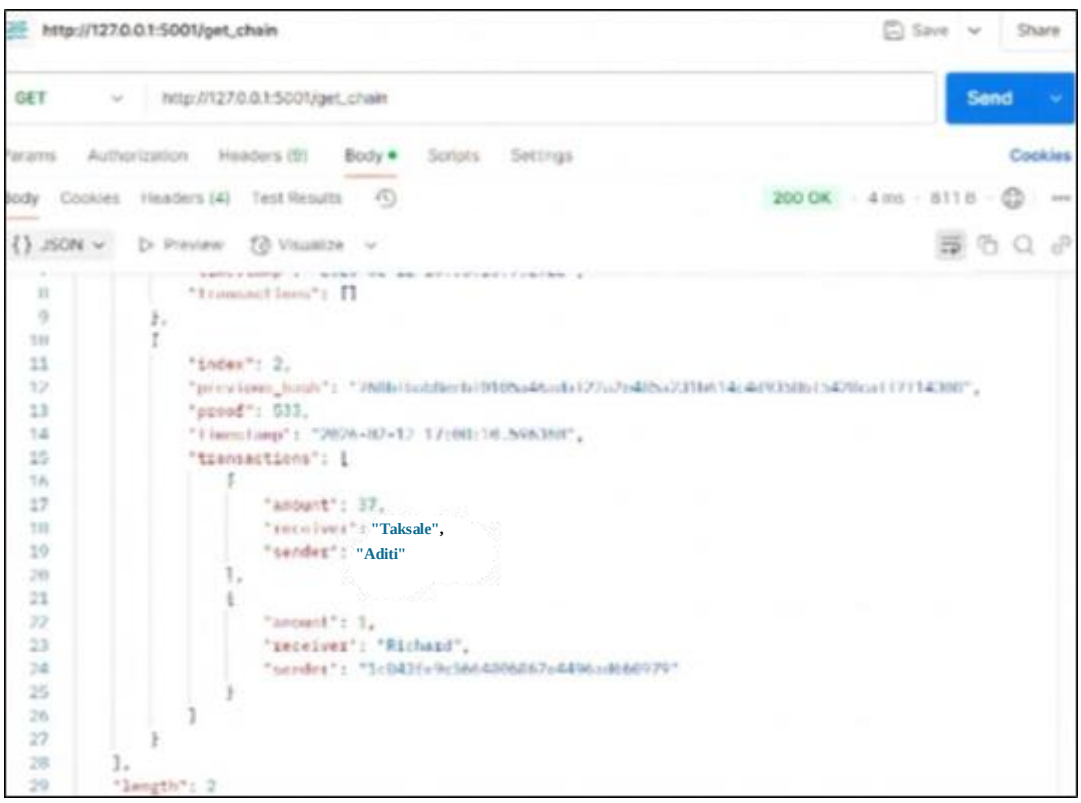
{} JSON Preview Visualize


```
{
    "chain": [
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-02-12 16:35:28.083070",
            "transactions": []
        }
    ],
    "length": 1
}
```

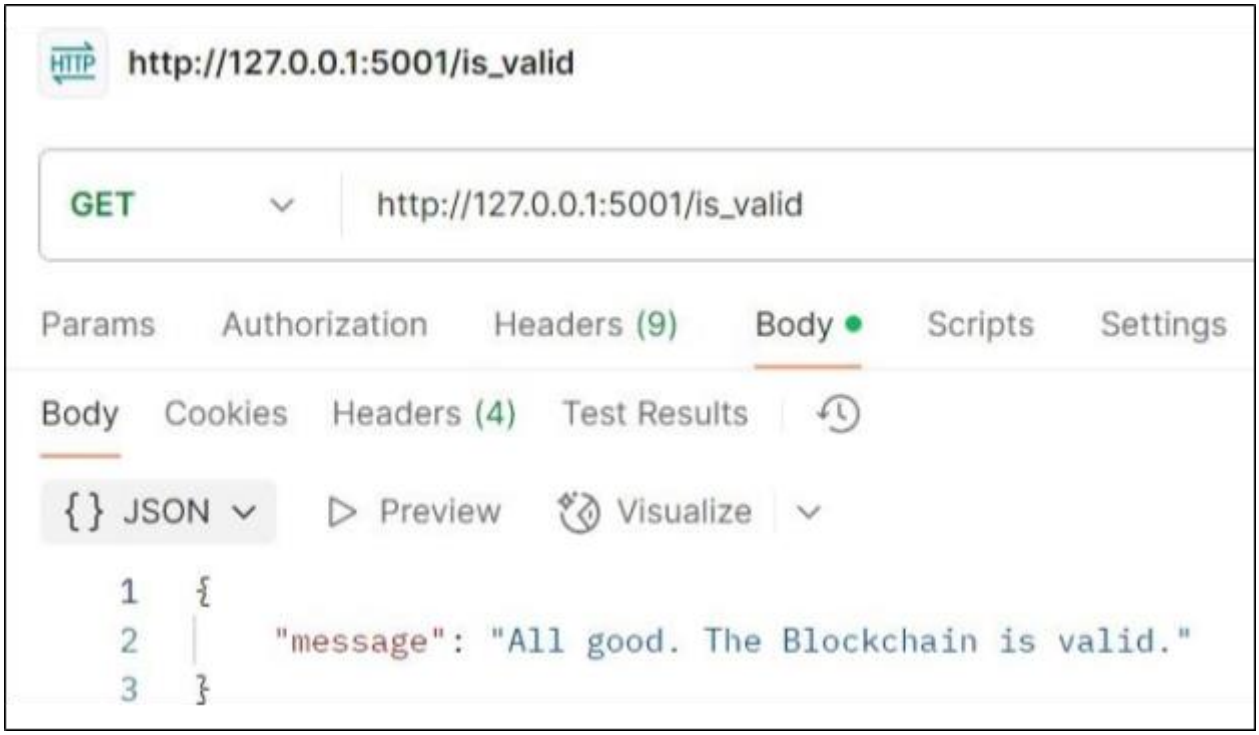
5. Replace Chain (Consensus Mechanism) : Node 2



Node 3



6. **Final Validation**



7. **Terminal Activity**

```
(venv) PS C:\Users\prajj\Downloads\Lab_3_Create a Cryptocurrency-20260211T182232Z-1-001\Lab_3_Create a Cryptocurrency> p
ython hadcoin_node_5001.py
>>
  File "C:\Users\prajj\Downloads\Lab_3_Create a Cryptocurrency-20260211T182232Z-1-001\Lab_3_Create a Cryptocurrency\hadc
oin_node_5001.py", line 164, in connect_node
    nodes = json.get('nodes')                                    # Get the nodes from json file
AttributeError: 'NoneType' object has no attribute 'get'
127.0.0.1 - - [12/Feb/2026 16:50:52] "POST /connect_node HTTP/1.1" 500 -
127.0.0.1 - - [12/Feb/2026 16:54:03] "POST /connect_node HTTP/1.1" 201 -
127.0.0.1 - - [12/Feb/2026 17:06:56] "POST /add_transaction HTTP/1.1" 201 -
127.0.0.1 - - [12/Feb/2026 17:07:55] "POST /mine_block HTTP/1.1" 405 -
127.0.0.1 - - [12/Feb/2026 17:08:10] "GET /mine_block HTTP/1.1" 200 -
127.0.0.1 - - [12/Feb/2026 17:11:22] "GET /get_chain HTTP/1.1" 200 -
127.0.0.1 - - [12/Feb/2026 17:15:11] "GET /get_chain HTTP/1.1" 200 -
127.0.0.1 - - [12/Feb/2026 17:16:11] "GET /get_chain HTTP/1.1" 200 -
127.0.0.1 - - [12/Feb/2026 17:17:08] "GET /is_valid HTTP/1.1" 200 -
```

Conclusion: In this experiment, we successfully created a basic cryptocurrency using Python and Flask and implemented mining in a multi-node blockchain network. We demonstrated how transactions are added, stored temporarily, and included in a block during the mining process using a Proof-of-Work mechanism. By running three separate nodes, we simulated a
peer-to-peer network and applied the Longest Chain Rule to maintain consensus

across all nodes. The experiment helped us understand blockchain structure, decentralized networking, mining rewards, and chain synchronization in a practical manner.